Chapter 1

1. Introduction :

What is AI ?

> "some consider **intelligence** to be a property of **internal thought processes and reasoning**, while others focus on **intelligent behavior, an external characterization**"

1. Acting humanly : The Turing Test Approach

Turing Test :

- classic method to evaluate a machine's ability to exhibit intelligence behavior indistinguishable from that of a human
- A computer passes the test if a human interrogator, after posing some written questions, cannot tell whether the written responses come from a person or from a computer.
  The computer must have some capabilities to pass the test :
- ***natural language processing*** : to communicate successfully in human language
- ***knowledge representation*** : storing what it knows or heard.
- ***Automated Reasoning***: to answer questions and draw new conclusions .
- ***machine learning*** : detect and predict and adapt to new circumstances .

Total Turing Test :

- another extended version of Turing test where the machine interacts with the objects around it , thus which will require the machine to have these additional capabilities :
- ***Computer vision*** : image and speech recognition to perceive the world
- ***Robotics*** : to manipulate objects and move around .

Limitations of the Turing test*

- **not Reproducible** : if we repeat the same test , we may not get the same results .
- **Not constructive** : does not have a metric for what constitutes intelligence . (..based on human subjective judgment)

2. Thinking humanly : The cognitive modeling approach

To tell if a machine thinks like a human we must know how humans think ? and there are three ways for that

- **introspection** : trying to catch our own thoughts as they go by .
- **Psychological experiments** : observing a person in action .
- **brain imaging**: observing the brain in action .

Cognitive Science :

The science that explores questions like: How do we recognize faces? Why do we forget things? How do we learn new skills? **Cognitive science** tries to answer these questions by looking at both the brain (biology) and the mind (mental processes).

- **goal** mimic how humans think is what ai is about

3. Thinking Rationally : The "Laws of thought" Approach

Logic :

- Aristotle started with his *syllogisms* , where he provided an argument structure that always yields correct conclusions.

  > *"Socrates is a man and all men are mortal and concludes that Socrates is mortal"*

- these laws initiated *logic* , then its precise notation for statements and objects surfaced then programs based on logic were able to solve problems .
- We're not always certain of the states of life , so there comes *probability* allowing reasoning with uncertain information .

  > *"In Probability principle, it allows the construction of a comprehensive model of rational thought, leading from raw perceptual information to an understanding of how the world works to predictions about the future.*

4. Acting Rationally: The rational Agent Approach

- **Agent** : an entity that can act .
- **Computer Agent** : they're expected to do more operate *autonomously* , *perceive their environment* , *adapt* , *stays the same w/o changing* , *create , pursue goals*.
- **rational Agent** : agent that works so as to achieve the best outcome (best expected outcome)
  - **Rationality** is about making decisions that align with the agent's goals, not necessarily about mimicking human thought processes.

The "Laws of Thought" Approach vs. Rational-Agent Approach

- The **"laws of thought" approach** focuses on making correct inferences (logical reasoning) as a way to achieve rationality.
  - While correct reasoning is important, it is only one way to act rationally.
  - Some rational actions (e.g., reflex actions like recoiling from a hot stove) do not involve deliberate reasoning.
- The **rational-agent approach** is broader and more general:
  - It includes inference as one mechanism but also encompasses other ways of acting rationally.
  - It is more **scientifically rigorous** because rationality can be mathematically defined and analyzed.

- For agents to be rational y7tajo same requirements that the machines need to have to pass the Turing test .

Beneficial Machines

Limitation of standard model :

- it assumes that a fully specified objective is given to the AI system but that is not true .
- It might work well for well-defined tasks (chess , shortest path problems) but will struggle in real-world scenarios
  Value Alignment Problem
- ensuring that AI objectives align with human values and preferences .(act ethically)
  New formulation
  a better approach involves AI systems that :
- recognize uncertainty about human objectives
- Act cautiously , seek permission , learn human preferences over time .
- Defer to human control when necessary .

The Foundations of Artificial Intelligence

Philosophy

- it explored interesting questions " Can formal rules derive valid conclusions?"
- **Key Contributions**:
  - **Aristotle**: Developed syllogisms for logical reasoning.
  - **Ramon Llull**: Created mechanical systems for reasoning.
  - **Descartes**: Introduced dualism (mind vs. matter).
  - **Empiricism** (Locke, Hume): Knowledge comes from sensory experience.
  - **Logical Positivism** (Vienna Circle): Knowledge is based on logical theories tied to observations.
  - **Confirmation Theory** (Carnap, Hempel): Quantified belief based on evidence.
- **Connection Between Knowledge and Action**:
  - **Aristotle**: Actions are justified by logical connections between goals and outcomes.
  - **Utilitarianism** (Bentham, Mill): Decisions should maximize overall utility.
  - **Deontological Ethics** (Kant): Actions are governed by universal moral rules, not outcomes.
- **AI Implications**:
  - Early AI systems (e.g., Newell and Simon's General Problem Solver) were inspired by Aristotle's goal-oriented reasoning.
  - Modern AI incorporates utility maximization and decision-making under uncertainty.

Mathematics

1. **Formal Logic**:
   - *George Boole*: Developed Boolean logic (1847).
   - *Gottlob Frege*: Created first-order logic (1879), which became central to AI and computation.
2. **Probability and Statistics**

- *Probability*: Developed to handle uncertainty (e.g., Cardano, Pascal, Bayes).
- *Bayes' Rule*: A key tool for updating beliefs with new evidence.
- *Statistics*: Emerged from probability and data analysis (e.g., Fisher).

3. **Computation and Algorithms**:
   - *Algorithms*: Ancient roots (e.g., Euclid's algorithm), formalized by Boole and others.
   - *Gödel's Incompleteness Theorem*: Showed limits to what can be proven or computed.
   - *Turing*: Defined computability with Turing machines and showed some problems are uncomputable.
4. **Tractability and NP-Completeness**:
   - *Tractability*: Problems are intractable if solving time grows exponentially with size.
   - *NP-Completeness*: Identifies problems likely to be intractable, shaping AI's focus on efficient algorithms.

Economics

- **Origins**: Economics began with Adam Smith's *Wealth of Nations* (1776), analyzing self-interest while considering morality (*The Theory of Moral Sentiments*, 1759).
- **Early Theories**: Arnauld (1662) introduced expected value; Bernoulli (1738) proposed diminishing marginal utility.
- **Utility & Game Theory**: Walras (1834–1910) expanded utility theory; von Neumann & Morgenstern (1944) developed game theory.
- **Decision Theory**: Studies preferences and uncertainty, with game theory addressing multi-agent interactions.
- **Operations Research**: Bellman (1957) formalized Markov decision processes for sequential decisions.
- **AI & Economics**: Early AI ignored complex decision-making; Simon (1916–2001) introduced *satisficing* (good-enough choices). AI now integrates decision-theoretic methods.

Neuroscience

- **Brain Processes Information**: Through electrochemical activities in neural networks, enabling thought, consciousness, and behavior.
- **Key Historical Milestones**:
  - *Localization of Function*: Paul Broca identified Broca's area (1861) for speech production.
  - *Neuron Discovery*: Camillo Golgi's staining technique (1873) allowed visualization of neurons, advanced by Santiago Ramón y Cajal.
  - *Electrochemical Basis*: Cognitive functions arise from neuron interactions.
- **Modern Neuroscience Tools**:
  - *EEG (1929)*: Measures brain electrical activity.
  - *fMRI*: Provides detailed images of brain activity linked to cognition.
  - *Optogenetics*: Controls and measures light-sensitive neurons.
  - *Brain-Machine Interfaces (BMIs)*: Enable brain-device communication, showing brain adaptability.

- **Brain Plasticity**: Ability to reorganize and remap functions after damage or integrate external devices.
- **Unanswered Questions**:
  - How individual memories are stored.
  - Mechanisms of higher-level cognitive functions.
- **Brain vs. Computers**:
  - Brains are slower but have greater storage and interconnectivity.
  - Raw computational power alone isn't enough; theoretical breakthroughs in understanding intelligence are needed.

"Neuroscience has made progress, but many mysteries about the brain remain."

Psychology

- **Origins**: Scientific psychology began with *Helmholtz* and *Wundt*, who studied vision and established experimental methods, including introspection.
- **Behaviorism**: Led by *John Watson*, it focused on observable stimuli and responses, excelling in animal studies but limited in explaining human cognition.
- **Cognitive Psychology**: Views the brain as an information-processing system. *Kenneth* **Craik** proposed that agents use internal representations to model reality, enabling reasoning and decision-making.
- **Cognitive Science**: Emerged in the 1950s, combining *psychology* and *computer modeling*. Key works by *Miller*, *Chomsky*, and *Newell* & *Simon* linked memory, language, and logic to information processing.
- **Human-Computer Interaction (HCI)**: *Doug Engelbart* championed ***intelligence augmentation (IA)***, using computers to enhance human capabilities, demonstrated in his groundbreaking 1968 demo.
- **Modern Perspective**: IA (human-centered) and AI (machine autonomy) are now seen as complementary, both essential for effective human-machine collaboration.

"psychology has evolved from introspection and behaviorism to cognitive science, integrating computer models and emphasizing the synergy between human intelligence and machine capabilities."

Computer engineering

1.2.6 Computer Engineering: **How to Build an Efficient Computer**

- **Origins**: Early computers like Colossus (UK), Z-3 (Germany), and ENIAC (USA) emerged during WWII.
- **Moore's Law**: Computing power doubled every 18 months until 2005, shifting to multi-core processors and parallelism.
- **AI Hardware**: GPUs, TPUs, and quantum computing accelerate AI tasks, with a 300,000-fold increase in AI training power (2012–2018).
- **Early Innovations**: Jacquard Loom (1805), Babbage's Analytical Engine (universal computation), and Ada Lovelace's visionary insights.
- **AI's Impact**: Pioneered time-sharing, GUIs, programming paradigms, and data structures in mainstream computing.

In short: Computer engineering has advanced from early mechanical devices to AI-optimized hardware, driven by Moore's Law and parallelism, with AI significantly influencing modern computing.

Control theory and cybernetics

- **Origins**:
  - **Ktesibios** (250 BCE) built the first self-regulating machine, a water clock with a feedback mechanism.
  - Other early examples include James Watt's steam engine governor and Cornelis Drebbel's thermostat.
- **Mathematical Foundations**:
  - **James Clerk Maxwell** (1868) initiated the mathematical theory of control systems.
  - **Norbert Wiener** (1948) developed control theory further, linking it to cognition and coining the term *cybernetics*.
  - Wiener, with Rosenblueth and Bigelow, viewed purposive behavior as minimizing "error" between current and goal states.
- **Cybernetics and Cognition**:
  - Wiener, McCulloch, Pitts, and von Neumann organized influential conferences exploring computational models of cognition.
  - **W. Ross Ashby** (UK) proposed that intelligence arises from homeostatic devices with feedback loops for stable, adaptive behavior.
- **Modern Control Theory**:
  - Focuses on **stochastic optimal control**, minimizing a cost function over time—similar to AI's goal of optimal behavior.
  - Uses calculus and matrix algebra for systems with continuous variables.
- **AI vs. Control Theory**:
  - Control theory focuses on continuous, mathematically describable systems.
  - AI uses logical inference and computation to address problems like language, vision, and symbolic planning, which fall outside traditional control theory.

sum up : Control theory and cybernetics explore how systems can self-regulate using feedback mechanisms. While control theory focuses on continuous systems and optimization, AI extends these ideas to address complex, symbolic problems, creating a distinct but complementary field.

Linguistics

- **1957**: B.F. Skinner published *Verbal Behavior*, explaining language learning through behaviorism.
- **Chomsky's Critique**: Noam Chomsky reviewed Skinner's book, arguing behaviorism couldn't explain how children create and understand new sentences they've never heard before.
- **Chomsky's Theory**: In *Syntactic Structures*, Chomsky proposed a theory based on syntax (inspired by Panini, c. 350 B.C.) that could explain language creativity and was precise enough to potentially be programmed into computers.
- **Linguistics and AI**: Modern linguistics and AI developed together, leading to **computational linguistics** or **natural language processing**.

- **Complexity of Language**: Researchers realized understanding language requires more than just sentence structure—it also needs context, subject matter, and meaning.
- **1960s Insight**: This understanding became widely recognized in the 1960s.
- **Knowledge Representation**: Early work in AI focused on how to store knowledge so computers could use it, influenced by linguistics and philosophy of language.